

# RUDIN CONTINUITY PROBLEMS

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ABSTRACT. This paper gives a sample set of solutions from Chapter 4 of Rudin's Introduction to Mathematical Analysis. It contains problems 13,

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## 1. PROBLEM 1

Let  $E$  be a dense subset of a metric space  $X$ , and let  $f$  be a uniformly continuous real function  $f : E \rightarrow \mathbb{R}$  defined on  $E$ .

**Lemma 1.1.** *For  $n \in \mathbb{N}$ , let  $V_n$  be an open ball around  $p \in X$  with radius  $\frac{1}{n}$ . That is,  $V_n = B(p, \frac{1}{n})$ . If  $p \notin E$ , then we can choose a point  $x_n \in V_n$  such that  $x_n \in E$ . Then, the sequence  $\{f(x_n)\}$  converges to a limit  $L$ .*

*Proof.* Since  $p \notin E$  and  $E$  is dense in  $X$ ,  $p$  is a limit point of  $E$ . Therefore, every open ball around  $p$  contains  $x \in E$ . Because sets  $V_n$  are open balls around  $p$ , we can always choose a point  $x_n \in V_n$  such that  $x_n \in E$ .

Since  $f$  is uniformly continuous, for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that for all  $p, q \in E$ ,

$$d_X(p, q) < \delta \implies |f(p) - f(q)| < \varepsilon.$$

Likewise, for every  $\frac{\varepsilon}{2}$ , there exists  $\delta_2 > 0$  such that

$$(1.1) \quad d_X(p, q) < \delta_2 \implies |f(p) - f(q)| < \frac{\varepsilon}{2}.$$

Let  $N = \text{ceiling}\left(\frac{1}{\delta_2}\right)$ , and let  $n > N$ . Then,

$$(1.2) \quad d_X(x_n, p) < \frac{1}{n}$$

$$(1.3) \quad < \frac{1}{N} = \frac{1}{\text{ceiling}\left(\frac{1}{\delta_2}\right)}$$

$$(1.4) \quad < \delta_2$$

Let  $a, b > N$ . Applying (1.1) on (1.4),

$$(1.5) \quad |f(x_a) - f(p)| < \frac{\varepsilon}{2}$$

$$(1.6) \quad |f(x_b) - f(p)| < \frac{\varepsilon}{2}$$

Applying the triangle inequality,

$$\begin{aligned} |f(x_a) - f(x_b)| &\leq |f(x_a) - f(p)| + |f(p) - f(x_b)| \\ &< 2\left(\frac{\varepsilon}{2}\right) \\ &< \varepsilon. \end{aligned}$$

Therefore, using  $N = \text{ceiling}\left(\frac{1}{\delta_2}\right)$ , for all  $\varepsilon > 0$ , we can always choose an  $N$  such that for  $a, b > N$ ,  $|f(x_a) - f(x_b)| < \varepsilon$ . Therefore,  $\{f(x_n)\}$  is a Cauchy sequence. Since the codomain of  $f$  is  $\mathbb{R}$  and that every Cauchy sequence in  $\mathbb{R}$  converges, we have  $\{f(x_n)\}$  converges to some limit  $L \in \mathbb{R}$ .  $\square$

**Lemma 1.2.** *If we follow the same construction as **Lemma 1.1**, we would always obtain the same limit  $\{f(x_n)\} \rightarrow L$  no matter which  $x_n$  are chosen in  $V_n$ .*

*Proof.* Let  $\{f(a_n)\}$  and  $\{f(b_n)\}$  be two different sequences constructed in the same manner as Lemma 1.1. Their limits are

$$A = \lim_{n \rightarrow \infty} f(a_n), \quad B = \lim_{n \rightarrow \infty} f(b_n).$$

Using the convergence of both  $\{f(a_n)\}$  and  $\{f(b_n)\}$ , we have

$$A - B = \lim_{n \rightarrow \infty} f(a_n) - f(b_n),$$

and by scalar multiplication,

$$|A - B| = \lim_{n \rightarrow \infty} |f(a_n) - f(b_n)|.$$

Since  $f$  is uniformly continuous, for every  $\varepsilon$ , there exists  $\delta > 0$  such that for all  $p, q \in E$ ,

$$(1.7) \quad d_X(p, q) < \delta \implies |f(p) - f(q)| < \varepsilon.$$

Let  $N = \text{ceiling}\left(\frac{2}{\delta}\right)$ , and let  $n > N$ . Then,

$$(1.8) \quad d_X(p, a_n) < \frac{1}{n}$$

$$(1.9) \quad < \frac{1}{N} = \frac{1}{\text{ceiling}\left(\frac{2}{\delta}\right)}$$

$$(1.10) \quad < \frac{1}{\frac{\delta}{2}}$$

$$(1.11) \quad < \frac{\delta}{2}.$$

Applying (1.11) and the triangle inequality

$$\begin{aligned} d_X(a_n, b_n) &\leq d_X(a_n, p) + d_X(p, b_n) \\ &< \frac{\delta}{2} + \frac{\delta}{2} \\ &< \delta. \end{aligned}$$

Applying (1.12),

$$|f(a_n) - f(b_n)| < \varepsilon.$$

Therefore, using  $N = \text{ceiling}(\frac{2}{\delta})$ , for all  $\varepsilon > 0$ , we can choose an  $N$  such that for all  $n > N$ ,  $|f(a_n) - f(b_n)| < \varepsilon$ . This is equivalent to

$$|A - B| = \lim_{n \rightarrow \infty} |f(a_n) - f(b_n)| = 0.$$

Hence,

$$|A - B| = 0 \implies A - B = 0 \implies A = B.$$

That is, the limits of  $\{f(a_n)\}$  and  $\{f(b_n)\}$  are the same. Therefore, no matter which sequence of  $x_n$  are chosen in  $V_n$ ,  $\{f(x_n)\}$  converges to the same limit.  $\square$

**Theorem 1.3.** Let  $g: X \rightarrow \mathbb{R}$  be a function such that for all  $p \in X$ ,

1. If  $p \in E$ , then  $g(p) = f(p)$ .
  2. If  $p \notin E$ , then  $g(p) = L_p$ , where  $L_p$  is the extension for all  $p \notin E$  defined uniquely by Lemma 1.1 and 1.2.
- Then  $g$  is a continuous extension of  $f$ .

*Proof.* Let  $p \in X$ . Since  $E$  is dense in  $X$ , we have three cases:

1.  $p \notin E$  and  $p$  is a limit point of  $E$ .
2.  $p \in E$  and  $p$  is a limit point of  $E$ .
3.  $p \in E$  and  $p$  is not a limit point of  $E$ .

We will consider each case separately. First, we consider case (1).

According to Lemma 1.2, for every  $\varepsilon$ , there exists  $\delta_A$  such that for all  $x \in E$  and  $d_X(p, x) < \delta_A$ ,

$$(1.12) \quad |L_p - f(x)| < \varepsilon$$

$$(1.13) \quad \implies |g(p) - g(x)| < \varepsilon$$

Thus, for every  $\varepsilon$ , there exists  $\delta_L$  such that for all  $x \in E$  and  $d_X(p, x) < \delta_L$ ,

$$(1.14) \quad |L_p - f(x)| < \frac{\varepsilon}{3}$$

$$(1.15) \quad \implies |g(p) - g(x)| < \frac{\varepsilon}{3}$$

Applying the uniform continuity of  $f$ , for every  $\varepsilon$ , there exists  $\delta_C$  such that for all  $x, y \in E$  such that  $d_X(x, y) < \delta_C$ ,

$$(1.16) \quad |f(y) - f(x)| < \frac{\varepsilon}{3}$$

$$(1.17) \quad \implies |g(y) - g(x)| < \frac{\varepsilon}{3}$$

Let  $x$  be such that  $d_X(x, p) < \min(\delta_A, \delta_C)$ . If  $x \notin E$ , since  $E$  is dense in  $X$ ,  $x$  is a limit point of  $E$ . According to Lemma 1.2, for every  $\varepsilon$ , there exists  $\delta_X$  such that for all  $x' \in E$  such that  $d_X(x, x') < \delta_X$ ,

$$(1.18) \quad |L_x - f(x')| < \frac{\varepsilon}{3}$$

$$(1.19) \quad \implies |g(x) - g(x')| < \frac{\varepsilon}{3}$$

Choose  $x', p' \in E$  with the conditions that

$$x' \in B\left(x, \min\left[\frac{\delta_C - d_X(p, x)}{4}, \delta_L, \delta_X\right]\right), \quad p' \in B\left(p, \min\left[\frac{\delta_C - d_X(p, x)}{4}, \delta_L\right]\right).$$

Applying the triangle inequality,

$$\begin{aligned} d_X(x', p') &\leq d_X(x', x) + d_X(x, p) + d_X(p', p) \\ &\leq 2\left(\frac{\delta_C - d_X(p, x)}{4}\right) + d_X(x, p) \\ &\leq \frac{1}{2}\delta_C + \frac{1}{2}d_X(p, x) \\ &< \delta_C. \end{aligned}$$

Applying (1.17),

$$|g(x') - g(p')| < \frac{\varepsilon}{3}.$$

In addition, since  $d_X(x', x) < \delta_X$  and  $d_X(p', p) < \delta_L$ , applying (1.19) and (1.15),

$$|g(x') - g(x)| < \frac{\varepsilon}{3}, \quad |g(p') - g(p)| < \frac{\varepsilon}{3}.$$

Applying the triangle inequality,

$$(1.20) \quad |g(p) - g(x)| \leq |g(x') - g(x)| + |g(x') - g(p')| + |g(p') - g(p)|$$

$$(1.21) \quad < 3 \left( \frac{\varepsilon}{3} \right)$$

$$(1.22) \quad < \varepsilon.$$

If  $x \in E$ , then since  $d_X(p, x) < \delta_A$ , applying (1.13),

$$|g(p) - g(x)| < \varepsilon.$$

Therefore, for all  $x \in E$  and  $x \notin E$ , if  $d_X(x, p) < \min(\delta_A, \delta_C)$ , then  $|g(p) - g(x)| < \varepsilon$ . Therefore, applying  $\delta = \min(\delta_A, \delta_C)$ , for every  $\varepsilon$ , there exists  $\delta$  for all  $x \in X$  such that  $d_X(p, x) < \delta$ ,  $|g(p) - g(x)| < \varepsilon$ . Thus,  $g$  is continuous at  $p$ .

Next, we consider case (2). This is similar to case (1), so (1)'s logic can be followed closely. Instead of obtaining  $|g(p') - g(p)| < \frac{\varepsilon}{3}$ , using  $\delta_L$ , we use  $\delta_C$ . More precisely, we choose  $p'$  by

$$p' \in B \left( p, \min \left[ \frac{\delta_C - d_X(p, x)}{4}, \delta_C \right] \right),$$

so  $d_X(p, p') < \delta_C$ , so  $|g(p') - g(p)| < \frac{\varepsilon}{3}$  by using (1.20). The rest of the logic follows the same way, so  $g$  is continuous at  $p$ .

Next, we consider case (3). If  $p$  is not a limit point of  $E$ , then there exists a  $\delta$ , such that  $B(p, \delta)$  contains no points  $q \in E$  apart from  $p$ . Consider the ball  $V_{\text{small}} = B(p, \frac{\delta}{2})$ . Suppose  $q \neq p$ , and  $q \in B(p, \frac{\delta}{2})$ . Then,  $q \notin E$ , so  $q$  is a limit point of  $E$  since  $E$  is dense. Construct a ball of radius  $\frac{\delta}{4}$  around  $q$ . Then, there must be some  $q'$  in this ball i.e.

$$d_X(q', q) < \frac{\delta}{4},$$

so

$$\begin{aligned} d_X(p, q') &\leq d_X(p, q) + d_X(q, q') \\ &< \frac{\delta}{2} + \frac{\delta}{4} \\ &< \frac{3}{4}\delta \\ &< \delta, \end{aligned}$$

which is a contradiction because we supposed that  $B(p, \delta)$  contains no points  $q' \in E$ . Hence, there is no  $q \neq p$  in the ball  $B(p, \frac{\delta}{2})$ . For all  $\varepsilon > 0$ , choose  $\delta_2 = \frac{\delta}{2}$ , then  $d_X(x, p) < \delta_2 \implies d_Y(g(x), g(p)) < \varepsilon$ , so  $g$  is continuous at  $p$ .

Since  $g$  is continuous at points that fit cases (1), (2), and (3),  $g$  is continuous at every point  $p \in X$ . Thus, since  $g(p) = f(p)$  for all  $p \in E$ ,  $g$  is a continuous extension of  $f$ .  $\square$

## 2. PROBLEM 2

Let  $I = [0, 1]$  be the closed unit interval. Let  $f$  be a continuous mapping from  $I$  to  $I$ .

**Lemma 2.1.** *Let  $y$  be a limit point of  $E_+ = \{x : f(x) > x\}$ . Then  $f(y) \geq y$ .*

*Proof.* Let  $g : I \rightarrow [-1, 1]$  be defined as

$$g(x) = f(x) - x$$

Then, it is sufficient to prove that if  $y$  is a limit point of  $E_+ = \{x : g(x) > 0\}$ , then  $g(y) \geq 0$ . Applying that continuity is conserved under basic algebraic operations, since both  $f(x)$  and  $x$  are continuous,  $g(x) = f(x) - x$  is also continuous.

For contradiction purposes, suppose

$$(2.1) \quad g(y) < 0.$$

Applying the continuity of  $g$ , for  $\varepsilon = -\frac{g(y)}{2} > 0$ , there exists  $\delta$  such that for all  $x$ ,

$$|y - x| < \delta \implies |g(y) - g(x)| < -\frac{g(y)}{2}.$$

We will separate into cases. First, assume that  $g(x) \leq g(y)$ . Then,

$$g(x) \leq g(y) < 0 \implies g(x) < 0.$$

Since  $y$  is a limit point of  $E_+$ , there exists  $x$  in  $|y - x| < \delta$  such that  $x \in E_+$ . From the continuity condition,  $g(x) > 0$ . This is a contradiction.

Second, assume that  $g(y) < g(x)$ . Then,

$$\begin{aligned} |g(y) - g(x)| &= g(x) - g(y) < -\frac{g(y)}{2} \\ \implies g(x) &< \frac{g(y)}{2} < 0. \end{aligned}$$

Since  $y$  is a limit point of  $E_+$ , there exists  $x$  in  $|y - x| < \delta$  such that  $x \in E_+$ . From the continuity condition,  $g(x) > 0$ . This is a contradiction. In both cases, (2.1) cannot be true. Therefore, if  $y$  is a limit point of  $E_+$ ,

$$g(y) \geq 0.$$

□

**Lemma 2.2.** *Let  $y$  be a limit point of  $E_- = \{x : f(x) < x\}$ . Then  $f(y) \leq y$ .*

*Proof.* Proof structure followed exactly like Lemma 2.1. □

**Theorem 2.3.** *There exists  $x \in I$  such that  $f(x) = x$ .*

*Proof.* We will use a proof by contradiction, so we assume that

$$(2.2) \quad f(x) \neq x \text{ for all } x \in I.$$

Since  $f(I) \subseteq I$ , we have

$$f(0) \geq 0$$

and

$$f(1) \leq 1.$$

Applying (2.2),

$$(2.3) \quad f(0) > 0$$

$$(2.4) \quad f(1) < 1.$$

Applying (2.3), the set  $E_+ = \{x : f(x) > x\}$  is non-empty. Applying (2.4), the set  $E_- = \{x : f(x) < x\}$  is non-empty.

Applying (2.2), we have

$$f(I) = E_+ \cup E_-.$$

Applying Lemma 2.1 and Lemma 2.2, we have that the sets  $E_+$  and  $E_-$  are separated since they cannot contain each other's limit points. Since  $f(I)$  may be written as the union of two separated sets,  $f(I)$  is not connected. However, since  $I$  is connected and  $f$  is continuous, Rudin Theorem 4.22 implies that  $f(I)$  is connected. This is a contradiction. Hence, (2.2) is false:

$$f(x) = x \text{ for some } x \in I.$$

□

## 3. PROBLEM 3

A mapping of  $X$  into  $Y$  is considered **open** if  $f(V)$  is an open set in  $Y$  whenever  $V$  is an open set in  $X$ .

**Theorem 3.1.** *Every continuous open mapping from  $\mathbb{R}$  to  $\mathbb{R}$  is monotonic.*

*Proof.* It is sufficient to prove two cases:

1. If  $f$  is not monotonic and continuous, then  $f$  is not an open map.
2. If  $f$  is not monotonic and an open map, then  $f$  is not continuous.

Suppose  $f$  is not monotonic and  $f$  is continuous. Applying the fact that  $f$  is not monotonic, there exists a real open interval  $I = [a, b]$  such that

$$f(c) > f(b) \text{ and } f(c) > f(a)$$

for some  $c \in (a, b)$ . [Alternatively, we can also choose

$$f(c) < f(b) \text{ and } f(c) < f(a)$$

and obtain a similar proof using  $z = \inf(f(I))$ ].

Applying a continuous function  $f$ , the image  $f(I)$  of a compact set  $I$  is compact. Applying the Heine-Borel Theorem to  $f(I)$ ,  $f(I)$  is bounded, and thus has a least upper bound  $y = \sup\{f(I)\}$ . Since  $f(I)$  is compact,  $f(I)$  is closed, so it contains its limit points including  $y$ . Therefore, there exists  $x \in [a, b]$  such that  $f(x) = y$ . Since there exists  $c$  such that  $f(c) > f(a)$  and  $f(c) > f(b)$ , we have  $y \geq f(c)$ , so

$$f(x) > f(a) \text{ and } f(x) > f(b).$$

Therefore,  $x \neq a$  and  $x \neq b$ . Therefore,  $x \in (a, b)$ .

Let  $f(V)$  the image of the open set  $V = (a, b)$ . From above, we know that it has a maximum,  $y$ , such that  $y \in f(V)$ . Let  $Y(\varepsilon) = (y - \varepsilon, y + \varepsilon)$  be an open neighbourhood around  $y$  for  $\varepsilon > 0$ . Since  $y$  is the maximum,  $y + \varepsilon \notin f(V)$  for all  $\varepsilon > 0$ . Therefore, for all  $\varepsilon > 0$ ,  $Y(\varepsilon) \not\subset f(V)$ . Even though  $y \in f(V)$ ,  $y$  is not an interior point of  $f(V)$ , so  $f(V)$  is not open. Therefore,  $f$  is not an open mapping since there exists an open set,  $(a, b)$ , whose image is not open.

Suppose  $f$  is not monotonic and  $f$  is an open mapping. Applying the fact that  $f$  is not monotonic, there exists a real open interval  $I = (a, b)$ , such that

$$f(c) > f(b) \text{ and } f(c) > f(a)$$

for some  $c \in (a, b)$ . Since  $I$  is open,  $f(I)$  is an open set. If  $f$  is bounded above on  $(a, b)$ , then from the previous case, we know that  $f$  is not an open mapping. Therefore,  $f$  is unbounded on  $(a, b)$ . Extending to  $[a, b]$ ,  $f$  is also unbounded on  $[a, b]$ . Applying the Heine-Borel Theorem,  $f([a, b])$  is not compact. Since  $[a, b]$  is compact, if  $f$  is continuous, then  $f([a, b])$  is compact. Since  $f([a, b])$  is not compact,  $f$  is not continuous.

Since we have shown that if  $f$  is not monotonic, it cannot both be continuous and an open mapping. Therefore, if  $f$  is continuous and an open mapping, then  $f$  is monotonic.

□

## 4. PROBLEM 4

Every rational  $x$  can be written in the form  $x = m/n$ , where  $m$  and  $n$  are integers without any common divisors. When  $x = 0$ ,  $n = 1$ . Let  $f$  be defined as

$$f(x) = \begin{cases} 0 & \text{if } x \notin \mathbb{Q} \\ \frac{1}{n} & \text{if } x = \frac{m}{n}. \end{cases}$$

**Theorem 4.1.** *At every irrational point,  $f$  is continuous.*

*Proof.* It is sufficient to prove that for every  $\varepsilon$ , there exists  $\delta$  such that, for all irrational  $e \notin \mathbb{Q}$ ,

$$|x - e| < \delta \implies |f(x) - f(e)| < \varepsilon.$$

Let us consider the set of points  $Q_M$  defined as

$$Q_M = \{x \in \mathbb{R} : x = \frac{n}{M} \text{ for some } n \in \mathbb{Z}\}.$$

From the archimedean property, it is easy to show that there exists  $n$  such that

$$\frac{n-1}{M} < e < \frac{n}{M}$$

for all irrational  $e \notin \mathbb{Q}$ . Furthermore, let us define the distance between  $e$  and  $Q_M$  as

$$d(e, Q_M) = \min \left[ \left| e - \frac{n-1}{M} \right|, \left| e - \frac{n}{M} \right| \right].$$

Finally, we construct our wonderful  $\delta_M$  as

$$\delta_M = \frac{1}{2} \min [Q_1, \dots, Q_M].$$

For now,

$$|x - e| < \delta_M \implies |f(x) - f(e)| \leq \frac{1}{M}.$$

It is relatively easy to prove that it is continuous at  $x$  by taking arbitrarily large  $M$ . □

**Theorem 4.2.** *At every rational point,  $f$  is discontinuous. This discontinuity is a simple discontinuity*

*Proof.* Applying something similar to Theorem 4.1 and you'll get something like values below  $\frac{1}{2n}$  in a certain  $\delta$  domain which contradicts the continuity condition. It would also be a simple discontinuity as you'll approach zero on both sides. □

## 5. PROBLEM 5

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  have the intermediate value property: if  $f(a) < c < f(b)$ , then  $f(x) = c$  for some  $x$  in  $(a, b)$ .

**Theorem 5.1.** *If, for every  $r \in \mathbb{Q}$ , that the set of all  $x$  with  $f(x) = r$  is closed, then  $f$  is continuous.*

*Proof.* For the purpose of contradiction, suppose that  $f$  is not continuous. Since  $f$  is not continuous, there exists a convergent subsequence  $\{x'_n\}$  that limits to  $x_0$ , yet there exists a  $\varepsilon$  such that for every  $N$ , there exists  $n$  such that  $|f(x'_n) - f(x_0)| \geq \varepsilon$ . There would hence be a subsequence  $\{x_n\}$  that either follows:

1.  $f(x_n) > f(x_0)$
2.  $f(x_n) < f(x_0)$

Since the rational numbers are dense, it is possible to take a subsequence  $\{x'_n\}$  such that  $f(x'_n) > r > f(x_0)$  [or  $f(x'_n) < r < f(x_0)$ ]. Applying the intermediate value property,  $f(t_n) = r$  for some  $t_n$  between  $x_0$  and  $x_n$ ; thus  $t_n \rightarrow x_0$ . Thus,  $f(x_0) = r$  because the set is closed. This is a contradiction to 1 and 2. Therefore,  $f$  is continuous.  $\square$

## 6. PROBLEM 6

**Definition 6.1.** If  $E$  is a nonempty subset of a metric space  $X$ , define the distance from  $x \in X$  to  $E$  as

$$\rho_E(x) = \inf_{z \in E} d(x, z).$$

**Theorem 6.2.** The distance  $\rho_E(x) = 0$  if and only if  $x \in \overline{E}$ .

*Proof.* Suppose  $\rho_E(x) = 0$ . There are two cases:

1. There exists  $z \in E$  such that  $d(x, z) = 0$ .
2. For every  $\varepsilon$ , there exists  $z \in E$  such that  $d(x, z) < \varepsilon$ .

If (1) is true, then, from the definition of  $d$ ,  $z = x$ , so  $x \in E$ , and thus  $x \in \overline{E}$ . If (2) is true, then  $x$  is a limit point of  $E$ . Therefore,  $x \in \overline{E}$ .

If  $x \in \overline{E}$ , then  $x$  is either:

1.  $x \in E$ .
2.  $x$  is a limit point of  $E$ .

If  $x \in E$ , then choosing  $z = x$ ,  $d(x, x) = 0$ , so

$$0 \leq \inf_{z \in E} d(x, z) \leq 0 \implies \rho_E(x) = 0.$$

If  $x$  is a limit point of  $E$ , then for every  $\varepsilon$ , there exists  $z \in E$  such that  $d(z, x) < \varepsilon$ . □

**Theorem 6.3.** For all  $x, y \in X$ ,

$$|\rho_E(y) - \rho_E(x)| \leq d(x, y).$$

*Proof.* Without loss of generality, assume  $\rho_E(x) \leq \rho_E(y)$ , so

$$(6.1) \quad |\rho_E(y) - \rho_E(x)| = \rho_E(y) - \rho_E(x).$$

For every  $\varepsilon > 0$ , there exists  $z \in E$  such that  $d(x, z) - \rho_E(x) < \varepsilon$ . Therefore,

$$(6.2) \quad d(x, z) < \rho_E(x) + \varepsilon.$$

Moreover, we have

$$\begin{aligned} \rho_E(y) &\leq d(y, z) \\ &\leq d(x, y) + d(x, z) \\ &< d(x, y) + \varepsilon + \rho_E(x). \end{aligned}$$

It can be shown

$$\rho_E(y) \leq d(x, y) + \rho_E(x).$$

Applying (6.1),

$$|\rho_E(y) - \rho_E(x)| \leq d(x, y).$$

□

**Theorem 6.4.** The function  $\rho_E(x)$  is uniformly continuous.

*Proof.* It is sufficient to show that for every  $\varepsilon > 0$ , there exists  $\delta$  such that  $d(x, y) < \delta \implies |\rho_E(x) - \rho_E(y)| < \varepsilon$  for all  $x, y \in X$ . Choose  $\delta = \varepsilon$ . Then,

$$d(x, y) < \varepsilon \implies |\rho_E(y) - \rho_E(x)| \leq d(x, y) < \varepsilon,$$

by applying Theorem 6.3. □

## 7. PROBLEM 7

**Theorem 7.1.** *Let  $K$  and  $F$  be disjoint sets in a metric space  $X$ , where  $K$  is compact and  $F$  is closed. Then there exists  $\delta > 0$  such that  $d(p, q) > \delta$  if  $p \in K$  and  $q \in F$ .*

*Proof.* For the purpose of contradiction, suppose that  $K$  and  $F$  are disjoint, and that for every  $\delta > 0$ , there exists  $p \in K$  and  $q \in F$  such that  $d(p, q) \leq \delta$ . Let  $\delta_n = \frac{1}{n}$ , and choose  $p_n$  and  $q_n$  based on the above condition. Applying Rudin 3.6, there exists a subsequence of  $\{p_n\}$ ,  $\{p_{n_k}\}$ , that converges to a point  $p \in K$ .

Therefore, for every  $\varepsilon > 0$ , we can choose  $N_1$  such that for all  $k > N_1$ ,

$$d(p_{n_k}, p) < \frac{\varepsilon}{2}.$$

Since  $\delta_n \rightarrow 0$ , we may also choose for all  $N_2$  such that for  $k > N_2$ ,

$$d(p_{n_k}, q_{n_k}) < \frac{\varepsilon}{2}.$$

Applying the triangle inequality, for all  $k > \max(N_1, N_2)$ ,

$$d(q_{n_k}, p) < \varepsilon.$$

Therefore,  $\{q_{n_k}\} \rightarrow p$ . Since  $F$  is closed,  $p \in F$ . Therefore,  $p \in K$  and  $p \in F$ , so  $K$  and  $F$  are not disjoint. Since  $K$  and  $F$  are disjoint, this is a contradiction. Thus, if  $K$  and  $F$  are disjoint, then there exists  $\delta > 0$  such that  $d(p, q) > \delta$  if  $p \in K$  and  $q \in F$ .  $\square$

## 8. PROBLEM 8

Let  $A$  and  $B$  be disjoint nonempty closed sets in a metric space  $X$ . Let  $f : X \rightarrow [0, 1]$  be defined as

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)}.$$

This function will always exist since it is not possible for  $\rho_A(p) = \rho_B(p) = 0$ . It would also be continuous because any algebraic combination of continuous functions are continuous provided that the denominator never equals zero.

**Theorem 8.1.** *Setting*

$$V = f^{-1}([0, \frac{1}{2})) \quad W = f^{-1}((\frac{1}{2}, 1]),$$

*$V$  and  $W$  are open and disjoint.*

*Proof.* Disjoint is trivial.  $f^{-1}((0, \frac{1}{2}))$  is open since  $f$  is continuous.  $[0, \frac{1}{2})$  is open in  $[0, 1]$  anyway so yeah.  $\square$

## 9. PROBLEM 9

**Definition 9.1.** A real-valued function  $f$  defined in  $(a, b)$  is **convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y),$$

when  $x, y \in (a, b)$  and  $\lambda \in (0, 1)$ .

**Theorem 9.2.** *If a function  $f$  is convex, then  $f$  is continuous.*

*Proof.* It is sufficient to show that for every  $\varepsilon > 0$ , there exists  $\delta$  such that for all  $p \in (a, b)$ ,

$$|x - p| < \delta \implies |f(x) - f(p)| < \varepsilon.$$

Consider the interval  $(p - \delta, p + \delta)$ . Then, the convex condition across the interval is

$$\begin{aligned} f(\lambda(p - \delta) + (1 - \lambda)(p + \delta)) &\leq \lambda f(p - \delta) + (1 - \lambda)f(p + \delta), \\ f(-2\delta\lambda + p + \delta) &\leq \end{aligned}$$

Suppose  $f$  is discontinuous. Then, there exists a point  $p$  that is either a:

1. **simple discontinuity:**  $\lim_{x \rightarrow p+} f(x) = \lim_{x \rightarrow p-} f(x) \neq f(p)$
2. **other discontinuity**  $\lim_{x \rightarrow p+} f(x) \neq \lim_{x \rightarrow p-} f(x)$ .

Now, suppose  $p$  is a simple discontinuity. Then, we have two cases:

1. (a)  $f(p) > \lim_{x \rightarrow p} f(x)$ ,
1. (b)  $f(p) < \lim_{x \rightarrow p} f(x)$ .

For case 1. (a), there would be two points on either side of  $p$ ,  $p_1 < p$  and  $p_2 > p$  such that  $f(p_1) < f(p)$  and  $f(p_2) < f(p)$ . Drawing a line between them would break the convexity condition.

For case 1. (b), draw a line between  $p_1$  and  $p$ , and it may be shown that there would be points right of  $p_1$  that break the convexity condition. Now, suppose  $p$  is an **other discontinuity**, then WLOG assume  $\lim_{x \rightarrow p+} f(x) > \lim_{x \rightarrow p-} f(x)$ . Then, choose a point  $p_2$  and  $p_1$  and there would be points left of  $p_1$  that break the convexity condition. Thus, if  $f$  is not continuous, it is not convex. Applying contrapositive logical reasoning, if  $f$  is convex,  $f$  is continuous.  $\square$

**Theorem 9.3.** *Every increasing convex function of a convex function is convex.*

*Proof.* Let  $h = g(f(x))$ , where  $f$  is a convex function, and  $g$  is an increasing convex function. Then, it is sufficient to show that

$$\begin{aligned} h(\lambda x + (1 - \lambda)y) &\leq \lambda h(x) + (1 - \lambda)h(y). \\ g(f(\lambda x + (1 - \lambda)y)) &\leq \lambda g(f(x)) + (1 - \lambda)g(f(y)). \end{aligned}$$

Just separate the  $f$  and becomes  $g$  is increasing you get the inequality  $\square$

## 10. PROBLEM 10

**Theorem 10.1.** *Suppose  $X$ ,  $Y$  and  $Z$  are metric spaces, and  $Y$  is compact. Let there be the functions  $f: X \rightarrow Y$ , and a continuous one-to-one function  $g: Y \rightarrow Z$ , and let  $h = g(f(x))$  for  $x \in X$ . Then, if  $f$  is not uniformly continuous, then  $h$  is not uniformly continuous.*

*Proof.* Applying Rudin Theorem 4.19, since  $g$  is a continuous function whose domain is a compact space  $Y$ ,  $g$  is a uniformly continuous function. If  $f$  is not uniformly continuous, then there exists a  $\varepsilon > 0$  such that for all  $\delta$ , there exists  $p, q \in X$  such that  $d_X(p, q) < \delta$  and

$$d_Y(f(p), f(q)) \geq \varepsilon.$$

Since  $g$  is a uniformly continuous one-to-one function, for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that for all  $d_Y(p, q) < \delta$

$$d_Z(g(p), g(q)) < \varepsilon.$$

We should expect  $\delta$  to have an upper limit, since if  $\delta$  can be made arbitrarily large for all  $\varepsilon$ , then  $g$  would be a constant function, which is not allowed since it is a one-to-one function. Let us name the upper limit of every  $\delta(\varepsilon)$  as  $L[\delta(\varepsilon)]$ , which can be made arbitrarily small as  $\varepsilon \rightarrow 0$  [since if this were not true, then obtain some contradiction with previous statements]. Therefore, if we choose  $\varepsilon$  small enough so  $d_Y(p, q) < L[\delta(\varepsilon)]$ , which is not possible since  $f$  is not uniformly continuous, so we get  $h$  is not uniformly continuous.

Eaiser if you just prove  $f(x) = g^{-1}(h(x))$  i guess lol. □

**Corollary 10.2.** *If  $h$  is uniformly continuous, then  $f$  is uniformly continuous.*

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